INVESTMENT ASSESSMENT

BIOECONOMY · CARBON REMOVAL

Forest-Residue Biochar Pyrolysis Plant

Financial, development and risk assessment of a 45,360 dry-tonne-a-year slow pyrolysis plant producing biochar and certified carbon removal

33.8%	$38.9M	$19.1M	16,330 t
RETURN	NPV AT 10.5%	PROJECT COST	BIOCHAR A YEAR

Prepared by Ahmed Alayidi

October 10, 2026

Contents

	The investment scenario	2
	Key figures at a glance	3
	Executive summary	4
PART I · PROJECT AND MARKET		
1	Investment overview	6
2	Technology description	7
3	Market context	8
4	Development impact and monitoring framework	9
PART II · COSTS AND RETURNS		
5	Project cost and financing plan	11
6	Operating economics	12
7	Basis of the financial analysis	14
8	Financial analysis	15
9	Sensitivity and scenario analysis	17
PART III · RISK AND DELIVERY		
10	Risk assessment	21
11	Implementation arrangements	22
12	Safeguards and social due diligence	24
13	Recommendation and conditions	25
ANNEXES		
A	Assumption register	27
B	Model, method and checks	29
C	Sources	31

How to read this report

Every number we report in this assessment comes from one run of documented financial model. Each input is marked Sourced when we took it from a named source, with the page or table of that source, or Assumption when it is a planning assumption we made. We list the model files in Annex B.

The investment scenario

This report appraises **a plant that does not exist**. No sponsor, site owner, lender or buyer stands behind it. It is a desk study that asks one question: if a developer built one slow pyrolysis plant from a published engineering design and sold its biochar and carbon removal credits at today's market prices, **would the plant pay back its cost of capital?**

The plant is the **Worden, Oregon reference design** in Sessions et al. (2019), thermal pyrolysis with heat recovery only (their Scenario 2). Every other section of this report rests on **the givens below**.

Given	Value	Status	Source
Site and design	Worden, Oregon; thermal slow pyrolysis; heat recovery only, no electricity	Sourced	Sessions et al. 2019, p. 178, Table 3; p. 180, Table 5 (Scenario 2)
Feedstock	45,360 dry tonnes of forest residue a year at $55 a dry tonne delivered	Sourced	Sessions et al. 2019, p. 177, Section 2.2.1; p. 181, Section 3.5
Output	16,330 tonnes of biochar a year at full capacity	Sourced	Sessions et al. 2019, p. 184, Section 4
Capital	$19.1 million, including engineering and a 15% contingency	Sourced	Sessions et al. 2019, p. 180, Table 5
Production cost	$207 per tonne of biochar	Sourced	Sessions et al. 2019, p. 182, Table 9
Biochar price	$500 a tonne	Assumption	None; see Section 3
Credit price	$148 per tonne of CO2	Sourced	S&P Global, Oct. 2025, as reported by Carbon Credits, Dec. 3, 2025
Credits per tonne of biochar	2.5 tonnes of CO2, less 15% fees	Assumption	None; to be set by lab test and registry method
Financing	60% debt at 7.5%, 40% equity at 15.0%, giving a 10.5% cost of capital	Assumption	None; to be replaced by lender and sponsor terms
Timeline	1 construction year, 20 operating years; 60%, 85%, then full capacity	Assumption	Life: Sessions et al. 2019, p. 182, Table 9 header; start-up: none
Price basis	Flat nominal US dollars, held constant over the whole period	Assumption	None; the source study does not state the year of its dollars

Sources: Sessions et al. (2019), *Biomass and Bioenergy* 123: 175 to 185; S&P Global (Oct. 2025), as reported by Carbon Credits (Dec. 3, 2025). Full references in Annex C.

The analysis is a **project-level, pre-tax screening appraisal**. It leaves out income tax, depreciation, loan repayment, interest during construction, working capital, grants and non-market benefits. It is **not a bankable financial model**, and it would need lender term sheets, vendor quotes and a tax review before it could become one.

Key figures at a glance

33.8%
Financial internal rate of return

10.5%
Weighted average cost of capital (the hurdle)

$38.9M
Net present value at the cost of capital

$19.1M
Project cost, including engineering and a 15% contingency

3.1 yrs
Simple payback from start of operation

$192/t
Biochar price below which value turns negative

RECOMMENDATION

Proceed to development stage, and commit the **$19.1 million construction budget** only once biochar offtake is signed and the terms of the credits are confirmed, under the **five conditions** we set in Section 13.

Item	Value
Project	Slow pyrolysis plant converting forest residue to biochar
Reference design	Worden, Oregon site; thermal pyrolysis; heat recovery only (Scenario 2 in Sessions et al. 2019)
Feedstock	45,360 dry tonnes a year of forest thinning and logging residue
Products	16,330 tonnes a year of biochar; 40,824 tonnes of CO2 credited a year
Financing plan	60% debt, 40% equity (assumption)
Evaluation period	1 construction year plus 20 operating years
Price basis	US dollars, prices and costs held flat, pre-tax

In this report, "$" refers to US dollars, "t" refers to metric tonnes, and "CO2" refers to tonnes of carbon dioxide.

Executive summary

The plant clears its cost of capital by wide margin, namely a return of **33.8%** against hurdle of **10.5%**, and a net present value of **$38.9 million**.

Recommendation: **proceed to the development stage**, with **$19.1 million** construction commitment being conditional on **signed offtake of biochar** and on **confirmed credit terms**.

1 Biochar price decides the case

With credits, biochar must sell above **$192 a tonne** for the value to stay positive, while without credits, biochar must sell above **$507 a tonne**, which is just above the $500 we assume here. At $500 and no credits whatsoever, net present value is **minus $0.8 million**.

2 Standard adverse tests leave the project viable

A 10% fall in revenue lowers the return to **28.1%**, and one-year delay with 10% higher capital lowers the return to **25.0%**. Revenue can fall **37.8%** before the project stops covering the cost of capital of the project.

Return stays well above 10.5% hurdle under every standard test

FIGURE 1

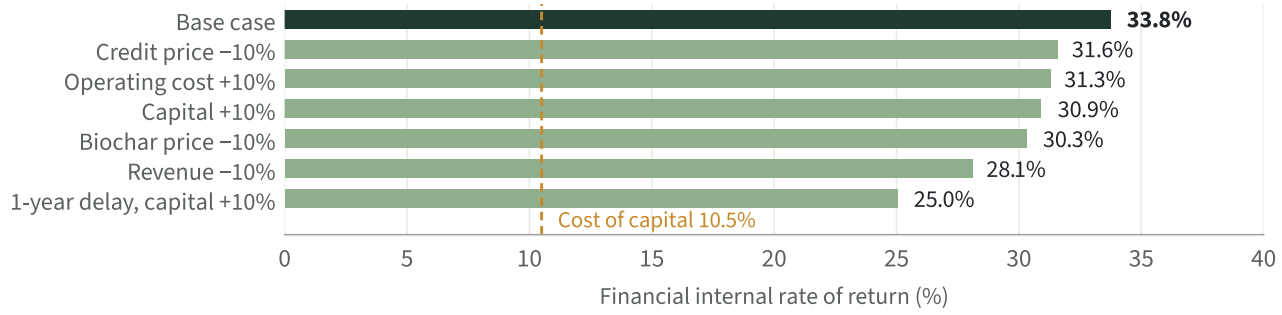


Figure 1. Financial internal rate of return in base case and under each one of the standard adverse tests. Source: model run.

Evidence base

The plant design, capital, yield, feedstock cost, production cost, life and salvage value come from **one reference design** in published engineering study of forest-residue biochar plants in Oregon and California. That study **does not state the year of its dollars**, and we did not adjust the figures of that study to today. Biochar price, credit volume, credit fees, start-up curve and financing terms are **assumptions we made in this assessment**.

Part I

Project and market

1	Investment overview	6
2	Technology description	7
3	Market context	8
4	Development impact and monitoring framework	9

Section 1 | Investment overview

Stand-alone slow pyrolysis plant takes in forest thinning and logging residue, and from that residue, it sells two products: **biochar and certified carbon removal credits**.

Item	Value	Source
Feedstock	Forest residue, 45,360 dry tonnes a year	Sessions et al. 2019, p. 177
Biochar yield	36% of dry feedstock weight	Sessions et al. 2019, p. 178, Table 2
Biochar output	16,330 tonnes a year at full capacity	Sessions et al. 2019, p. 184
Staff	26.5 full-time workers	Sessions et al. 2019, p. 181, Table 6
Operating life	20 years	Sessions et al. 2019, p. 182, Table 9
Start-up	60% of capacity in year 1, 85% in year 2, full from year 3	Assumption

The reference design is the **Worden, Oregon site with thermal pyrolysis and heat recovery only**, which Sessions et al. (2019) label **Scenario 2**. In that design, pyrolysis gases are burned in furnace whose heat dries the wood chips and drives the reactor, with no electricity whatsoever being generated.

Section 2 | Technology description

Slow pyrolysis heats dried biomass to several hundred degrees with little or no oxygen. The wood itself does not burn, and it breaks down instead into **solid, carbon-rich char**, plus gases and vapors that the plant burns in order to obtain its own heat.

Burning pyrolysis gases heats the dryer and the reactor, with biochar being the one product of the plant

FIGURE 2

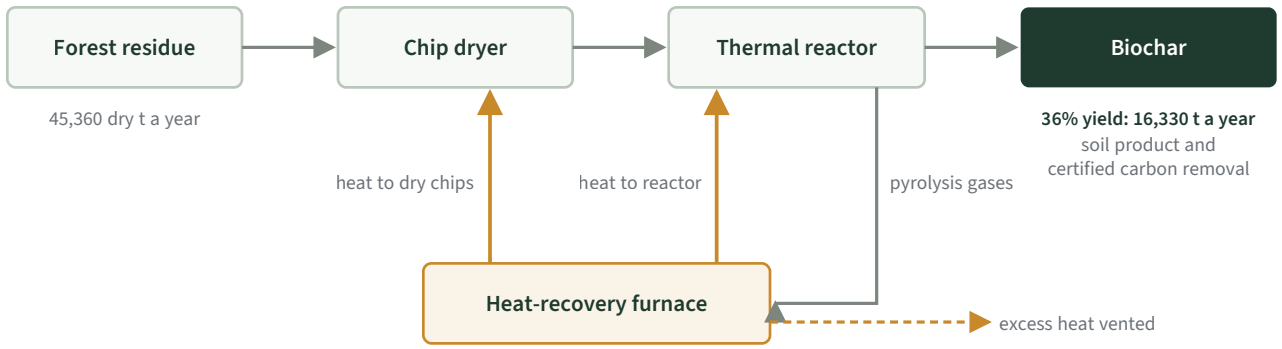


Figure 2. Process flow of reference design. Gray arrows show material flow, and amber arrows show heat flow. Source: Sessions et al. 2019, p. 178, Table 3.

We chose slow pyrolysis over fast pyrolysis given slow pyrolysis **maximizes the solid product**. Iowa State's Bioeconomy Institute reports char yields of **up to 40%** of feedstock weight for slow pyrolysis, against lower char yield for fast pyrolysis, with the latter favoring liquid bio-oil.

Carbon in biochar is **stable in soil for long periods**, which is why biochar can earn carbon removal credits. Adding to that, biochar is also sold as soil amendment for farms and nurseries.

Sessions et al. compared thermal and microwave pyrolysis at two sites under four energy-recovery designs. In the cost table of Sessions et al. (p. 182, Table 9), thermal pyrolysis at Worden with heat recovery only has **lowest total cost of the 16 cases**, namely **$446 per tonne** of biochar, covering feedstock, production and capital, and that is the reference design we use here.

Section 3 | Market context

Biochar is **volume leader in delivered carbon removal**, yet the demand for it is concentrated within **small number of large buyers**.

86%

Biochar share of delivered durable carbon removal credits (CDR.fyi, Sep. 2025)

$148/t

U.S. biochar credit price for 2026 delivery (S&P Global, Oct. 2025)

46%

Microsoft's share of biochar credits contracted (CDR.fyi, Sep. 2025)

Indicator	Value	Source
Biochar removal price corridor, base case	EUR 120 to 180 per tonne	Senken, Aug. 2026
Biochar removal market value, 2024	$181.5 million	CDR.fyi, Sep. 2025
Delivered as share of contracted, through Q2 2025	about 22%	CDR.fyi, Sep. 2025
Biochar minimum selling price, slow pyrolysis	$188 to $260 per tonne	NREL, 2025
U.S. biochar credit price, 2026 delivery	about $148 per tonne CO2	S&P Global, via Carbon Credits

Demand for credits is real but narrow, given top four buyers, which are Microsoft, Google, BCG and JPMorgan, account for **57% of biochar credit purchases**, and median buyer took **fewer than 100 tonnes** in the first half of 2025.

Price of the product itself

Price of the biochar product has **no single market benchmark**. Sessions et al. cite a 2014 survey average wholesale price of **$1,500 a tonne** among 28 U.S. producers (p. 183), and estimate that a plant of this size could produce biochar for **$500 to $600 a tonne**. In this assessment, we assume **sale price of $500 a tonne**, and we test a range going from **$250 to $700** for that price.

Section 4 | Development impact and monitoring framework

The outcome is plant running at **full capacity from operating year 3 onward**, turning 45,360 dry tonnes of forest residue a year into biochar and into certified carbon removal.

Capital and residue become running plant, which in turn delivers biochar and carbon removal

FIGURE 3

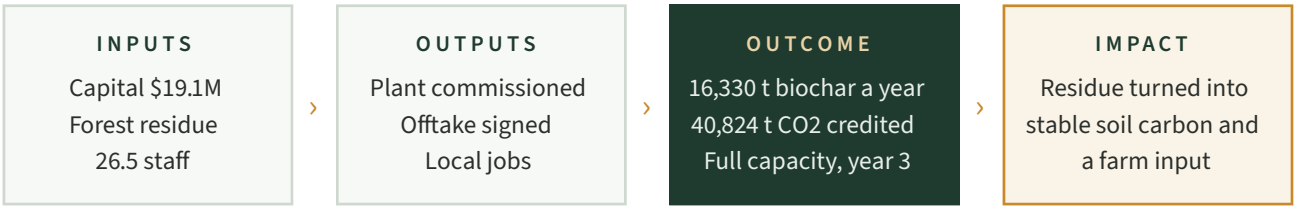


Figure 3. Results chain, following format of the ADB design and monitoring framework.

Results chain	Indicators, targets and baselines	Data sources	Risks and assumptions
Impact: forest residue converted into stable soil carbon and a farm input	Not measured at project level	Not applicable	Assumption: biochar and credit markets continue to grow
Outcome: biochar plant operating at full capacity	16,330 t of biochar a year by operating year 3 (baseline 0); 40,824 t CO2 credited a year (baseline 0); 45,360 dry t of residue processed a year (baseline 0)	Plant production records; registry issuance records	Risk: feedstock supply short of contract; credit buyers withdraw
Output 1: plant constructed and commissioned	Capacity of 45,360 dry t a year installed (baseline 0)	Commissioning certificate; supplier performance test	Risk: delayed commissioning
Output 2: offtake secured	Biochar and credit contracts covering at least 5 years signed before construction (baseline 0)	Signed contracts	Risk: buyers only commit short term
Output 3: local jobs created	26.5 full-time operating jobs (baseline 0)	Payroll records	Assumption: staff can be hired locally

Targets come from plant design in this assessment. Credited tonnage is 16,330 t of biochar times the 2.5 t CO2 per tonne we assumed.

Part II

Costs and returns

5	Project cost and financing plan	11
6	Operating economics	12
7	Basis of the financial analysis	14
8	Financial analysis	15
9	Sensitivity and scenario analysis	17

Section 5 | Project cost and financing plan

The project costs **$19.1 million**, which is financed **60% by debt and 40% by equity**, giving the project weighted average cost of capital of **10.5%**.

Reference design has the lowest total cost per tonne among the 16 designs that were studied

FIGURE 4

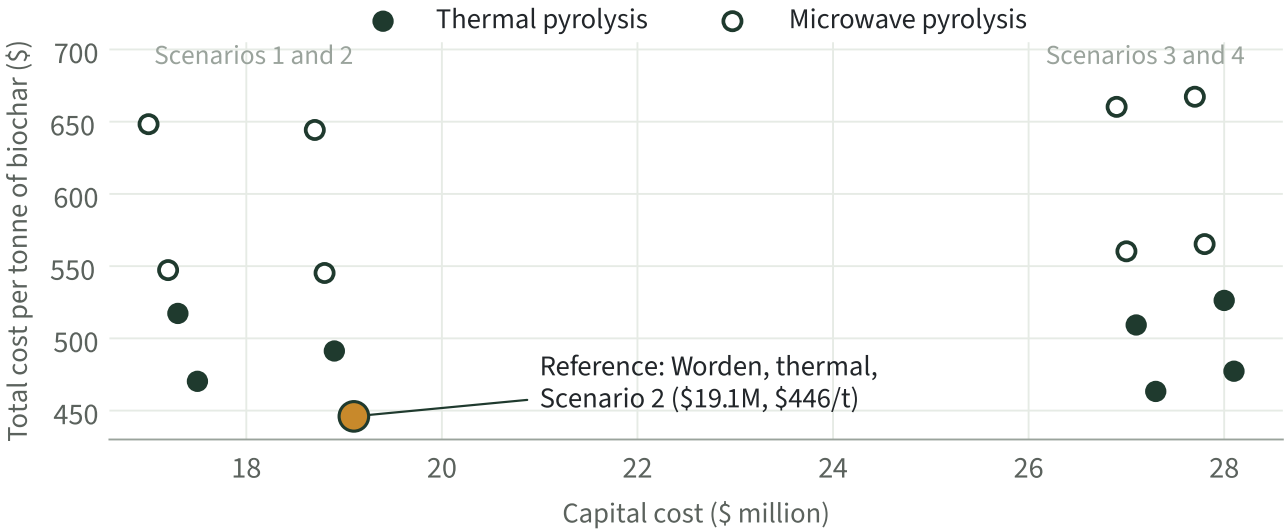


Figure 4. Capital cost against total cost per tonne of biochar (feedstock, production and capital) for every site, every technology and every energy-recovery design in Sessions et al. 2019 (p. 180, Table 5; p. 182, Table 9). Model input: designs.csv.

In the sensitivity tests, we use lowest capital of the 16 designs, **$17.0 million** (Yreka, microwave, Scenario 1), and the highest one, **$28.1 million** (Worden, thermal, Scenario 4). Capital estimates of the study include site preparation, buildings, equipment installation, engineering, contractor and owner costs, and a **15% contingency** (p. 180). The capital estimates do not mention working capital, and they are **not adjusted for the inflation of construction costs** since the study was published in 2019.

Weighted average cost of capital

Source	Amount ($ million)	Weight (%)	Nominal cost (%)	Composite cost (%)
Senior debt	11.46	60	7.5	4.5
Sponsor equity	7.64	40	15.0	6.0
Total	**19.10**	**100**		**10.5**

Debt share, debt rate and cost of equity are **assumptions we made**, to be replaced by lender term sheets. The cost of capital is pre-tax and nominal, matching the pre-tax cash flows of the project. We judge the project viable when its **financial internal rate of return exceeds this cost of capital**, which is the test ADB applies in its financial analysis for the Bihar Power System Improvement Project.

Section 6 | Operating economics

At full capacity, plant earns **$13.3 million** a year against **$5.9 million** of running cost, which leaves the plant with **$7.4 million** of operating cash.

Each full-capacity year turns $13.3M of revenue into $7.4M of operating cash for the plant

FIGURE 5

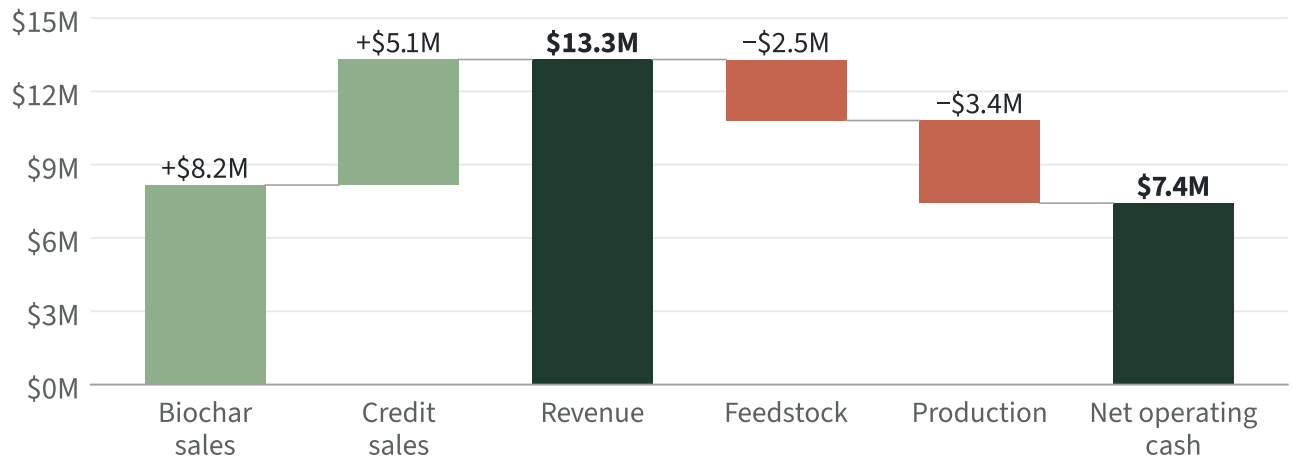


Figure 5. Revenue and operating cost in full-capacity year (operating year 3 onward), base case. Source: model run.

Biochar sales carry six-tenths of revenue, while production is larger cost of the two

FIGURE 6

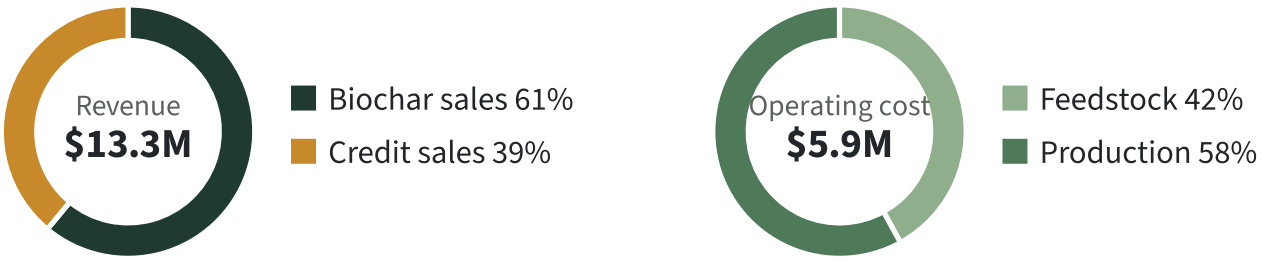


Figure 6. Shares of revenue and of operating cost in full-capacity year. Source: model run.

Each line is **quantity times price**, namely, biochar sales are 16,330 t at $500; credit sales are 16,330 t times 2.5 t CO2 per tonne at $148, less 15% fees; feedstock is 45,360 t at $55; and production is 16,330 t at $207. Biochar tonnage is 45,360 t times 36%, or 16,329.6 t, which we show rounded.

Feedstock: $55 a dry tonne

This is planning cost that Sessions et al. use for Worden site (p. 181). The range of the cost of collection and delivery to Worden in Sessions et al. runs from **$43 to $135 a tonne** (p. 179).

Production: $207 a tonne of biochar

This is the direct production cost the study reports for reference design (p. 182, Table 9), namely staffing, purchased energy, raw materials and other plant costs, excluding feedstock and capital. For thermal pyrolysis, the direct production cost ranges from **$184 to $254**.

Assumptions to confirm

Credit volume of **2.5 tonnes** of CO2 per tonne of biochar and the 15% fee share are assumptions that we made. The credited amount depends on the carbon content of the biochar and on the method of the registry, and it must be **confirmed by lab testing** and by the registry chosen.

Section 7 | Basis of the financial analysis

The analysis counts **only the cash flows the plant creates**, compared against without-project case in which no plant is built and no revenue or cost whatsoever arises.

Price basis

We hold all prices and costs flat in nominal US dollars over evaluation period, and we apply **no price escalation and no price contingency**. Source study **does not state the year of its dollars**.

Evaluation period

One construction year, then 20 operating years. Output starts at 60% of capacity in year 1, then goes to 85% in year 2, and to 100% of capacity from year 3 onward.

Salvage

20% of capital is recovered in operating year 20, as it is done in source study.

Tax

Cash flows are **before income tax**, and hence, we compare the cash flows with pre-tax cost of capital.

Financing

We exclude interest during construction and loan repayments from project cash flows, and financing enters the analysis **only through the cost of capital**.

Excluded

Depreciation, working capital, government grants, and non-market benefits such as reduction of wildfire hazard, all of which we leave out of the analysis.

Revenue

Biochar sales plus sales of carbon removal credits, net of 15% credit fees. We assume **every tonne produced to be sold** in the market.

Decision rule

The project is financially viable if its internal rate of return **exceeds the 10.5% cost of capital**, which is the same as positive net present value at that rate of **10.5%**.

Section 8 | Financial analysis

The financial internal rate of return is **33.8%**, against **10.5%** cost of capital, which supports financial viability of the project.

33.8%	$38.9M	$50.9M	$33.2M	3.1 yrs
Financial internal rate of return	NPV at 10.5%	NPV at 8% (rate used by Sessions et al.)	NPV at 12%	Simple payback from start of operation

Calculation of the financial internal rate of return ($ thousand)

Year	Biochar sales	Credit sales	Capital	Feedstock	Production	Salvage	Net benefit
0	0	0	19,100	0	0	0	(19,100)
1	4,899	3,081	0	1,497	2,028	0	4,455
2	6,940	4,365	0	2,121	2,873	0	6,312
3 to 19, each year	8,165	5,136	0	2,495	3,380	0	7,425
20	8,165	5,136	0	2,495	3,380	3,820	11,245
Total, years 0 to 20	**158,805**	**99,889**	**19,100**	**48,524**	**65,745**	**3,820**	**129,145**

( ) = negative. Figures may not add up exactly due to rounding of the figures. Credit sales are net of 15% fees. Source: model run, outputs/results.md.

FIGURE 7

Cumulative cash recovers $19.1M outlay in year 4

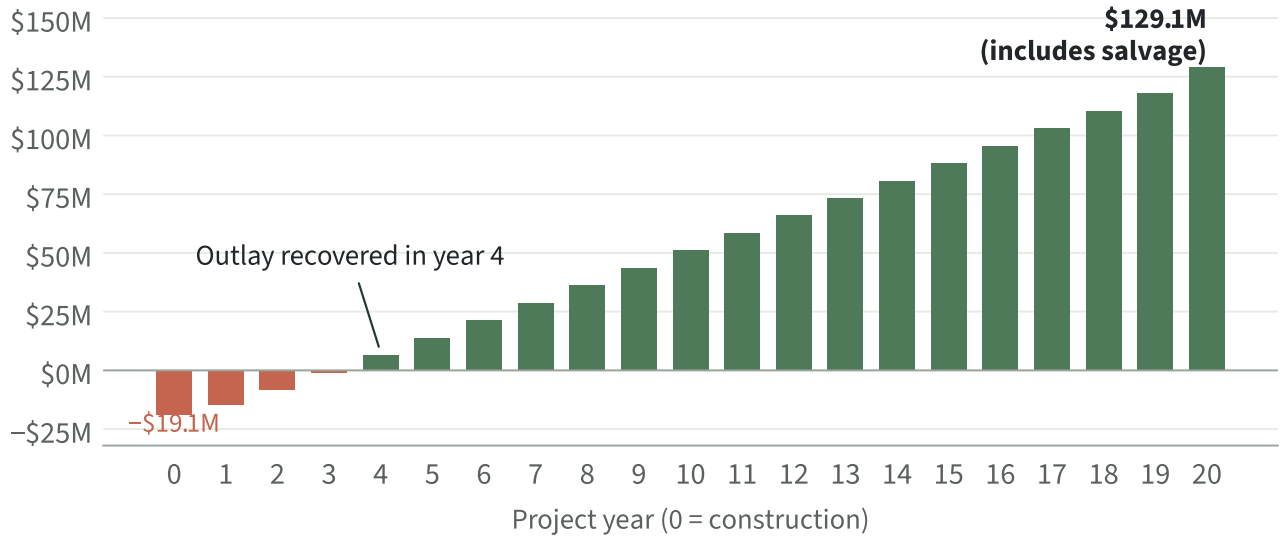


Figure 7. Cumulative pre-tax project cash flow, base case. Year 20 includes the salvage value of the plant. Source: model run.

The returns described above assume **every tonne of biochar sells at $500** and **earns credits at $148**. In Section 9, we show how fast the returns fall when either one of the two prices slips.

Section 9 | Sensitivity and scenario analysis

The return stays above **10.5%** cost of capital under **every standard adverse test**, and **revenue is the input the project is most sensitive to** among all inputs.

Results of financial sensitivity analyses

Sensitivity parameter	Variation	FIRR (%)	NPV at 10.5% ($ million)	Switching value (%)
Base case		33.8	38.9	
Revenue reduced	minus 10%	28.1	28.6	minus 37.8
Biochar price reduced	minus 10%	30.3	32.6	minus 61.6
Capital cost increased	plus 10%	30.9	37.1	plus 209.5
Operating cost increased	plus 10%	31.3	34.4	plus 85.6
Credit price reduced	minus 10%	31.6	35.0	minus 97.9
One-year delay and capital cost increased	1 year; plus 10%	25.0	31.6	
Weighted average cost of capital		**10.5**		

Switching value is the change in an input, on its own, that brings net present value at 10.5% down to zero. Operating cost covers feedstock and production.

Biochar sale price swings the value of the project more than any other input does

FIGURE 8

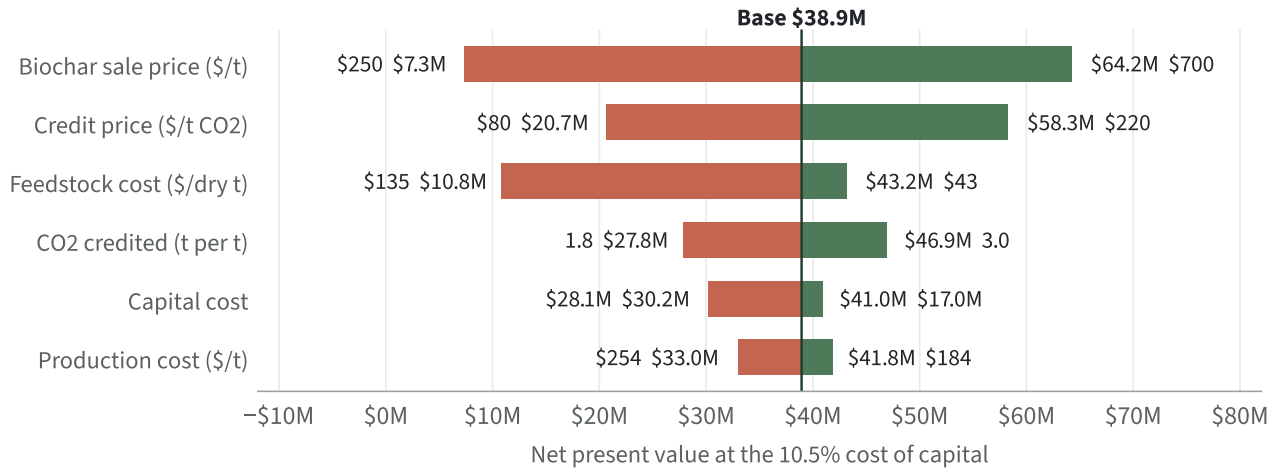


Figure 8. Net present value when one input moves to its low or high value and all other inputs stay at base. Ranges for capital, feedstock cost and production cost are from Sessions et al. (2019), while ranges for prices and credit volume are assumptions we made. Source: model run.

FIGURE 9

Biochar must sell above $192 a tonne with credits, or above $507 a tonne without credits

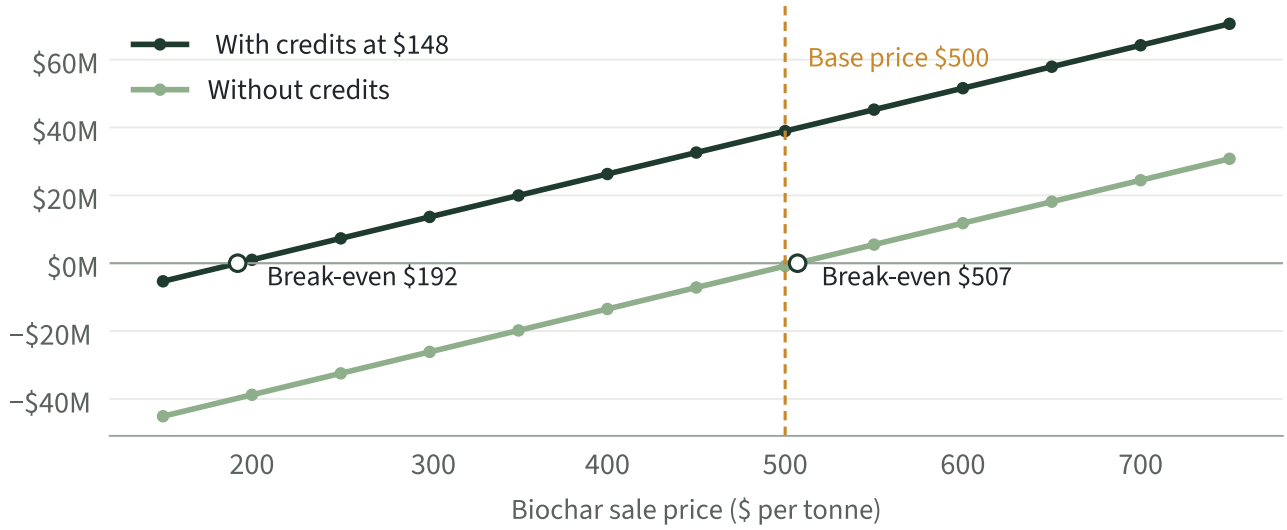


Figure 9. Net present value at the 10.5% cost of capital across biochar prices, with and without revenue from credits. Every $50 a tonne moves the value of the project by $6.3 million. Source: model run, price_curve in outputs/results.json.

$192	$507	$3	$166	$59.1M
Biochar price needed per tonne, with credits	Biochar price needed per tonne, without credits	Credit price needed per tonne CO2, at $500 biochar	Feedstock cost the project can carry per dry tonne	Capital the project can carry

FIGURE 10

Only combined downside, or losing all the revenue from credits, sinks the project

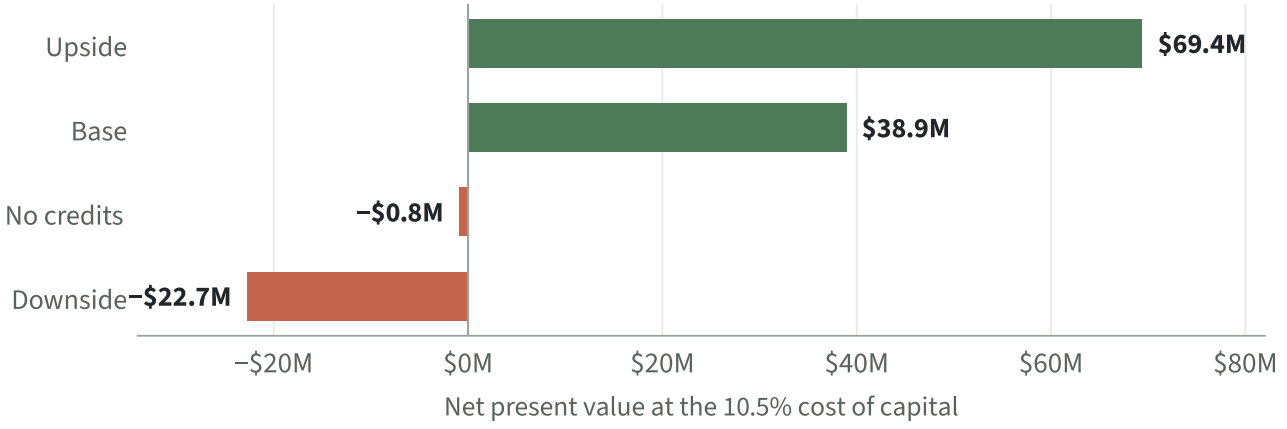


Figure 10. Net present value at 10.5% under the four scenarios we tested. Source: model run.

Scenario	Inputs	NPV at 10.5%	FIRR	Payback
Upside	Biochar $600, credits $180, capital $17.0M, production $184, feedstock $43	$69,355,459	53.9%	2.1 years
Base	As in the assumption register	$38,936,501	33.8%	3.1 years
No credits sold	Base, with zero credit revenue	minus $844,676	9.9%	8.9 years
Downside	Biochar $300, credits $100, capital $28.1M, production $254, feedstock $80	minus $22,748,275	minus 3.4%	Not reached

What to secure first

At $500 biochar, credit price of only **$3 a tonne** is enough to clear the cost of capital, while with no credit revenue whatsoever, the project **falls just short** of clearing the cost of capital. As such, **biochar sale price is the input to secure first**.



Risk and delivery

10	Risk assessment	21
11	Implementation arrangements	22
12	Safeguards and social due diligence	24
13	Recommendation and conditions	25

Section 10 | Risk assessment

We rate overall risk as **substantial**, driven by **market risks outside the control of the sponsor**, while project-specific risks are **moderate** and can be reduced by contract before construction starts.

External risks

Risk	Effect if it happens	Rating	Mitigation
Biochar sells below $192 a tonne	Net present value turns negative	High	Multi-year offtake with farm, nursery and landscaping distributors before financial close
Credit buyers pull back	At zero credit revenue, value falls to minus $0.8 million	High	Forward offtake for at least 5 years of credits; registry certification before start-up
Registry method credits less CO2 per tonne	At 1.8 t, value falls to $27.8 million	Medium	Confirm the registry method and lab-test biochar during design

Project-specific risks

Risk	Effect if it happens	Rating	Mitigation
Feedstock cost rises to $135 a tonne	Value falls to $10.8 million; break-even at $166	Medium	Long-term supply contracts with forest owners and land agencies; site near the residue source
Capital overrun	At $28.1 million, value is $30.2 million; break-even at $59.1 million	Low	Fixed-price engineering and construction contract; brownfield or sawmill site
Start-up delayed one year, capital up 10%	Return falls to 25.0%	Medium	Supplier performance guarantees on throughput and yield
Cost data from a 2019 study with no stated dollar year	Real costs may be higher today	Medium	Replace study figures with vendor quotes before approval

Ratings are the judgment of the assessor, based on how far each input can move the value, and the ratings are not probabilities.

Section 11 | Implementation arrangements

The plan runs from approval to full-capacity review at the end of operating year 3, which is **57 months after approval**, with construction held behind **financial close gate**.

Aspect	Arrangement
Sponsor and management	Open question: sponsor and operating company to be named
Implementation period	About 21 months from approval to commissioning
Construction	Fixed-price engineering, procurement and construction contract
Technology supplier	Conventional thermal slow pyrolysis with heat recovery, selected by competitive bid
Feedstock supply	Long-term contracts with forest owners and land agencies at about $55 a dry tonne delivered
Revenue structure	Biochar offtake contracts plus carbon removal credit offtake contracts, each covering at least 5 years
Operating cost structure	Feedstock 42% and production 58% of operating cost at full capacity
Staffing	26.5 full-time workers
Performance monitoring	Production and registry issuance reports against the indicators in the monitoring framework

Construction starts only after financial close gate at month 9

FIGURE 11

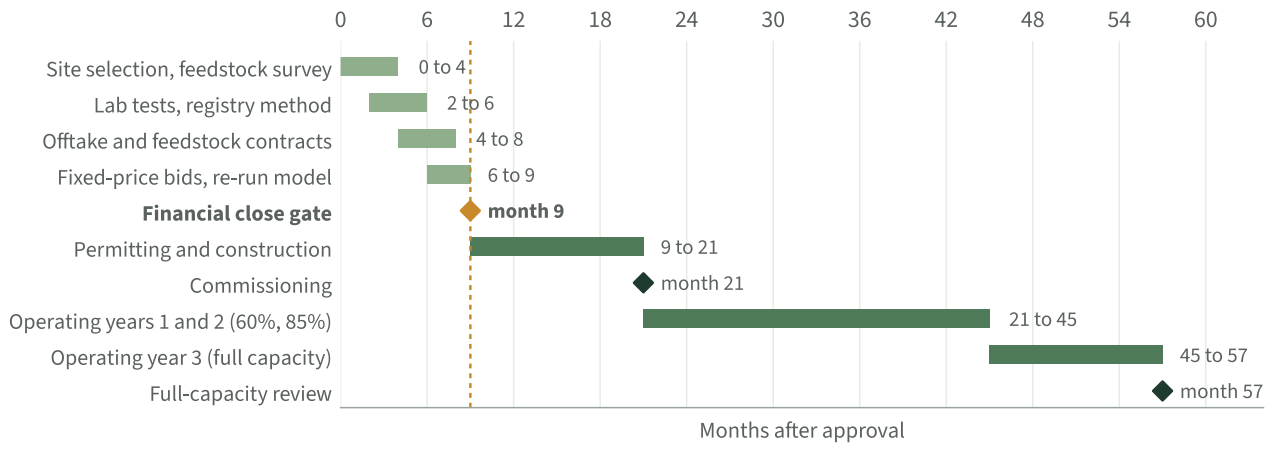


Figure 11. Plan from approval to the full-capacity review, in months after the approval. Durations are planning assumptions we made, to be confirmed with chosen technology supplier.

1 Site selection, with priority on brownfield or sawmill co-location, and survey of feedstock supply within haul distance of the site.

2 Lab tests of biochar made from the feedstock of the site, then confirm credited CO2 per tonne with chosen registry.

3 Sign offtake agreements for biochar and for credits, and sign contracts of feedstock supply with the suppliers.

4 Fixed-price engineering and construction bids, replacing cost figures of the study with the quotes received, and re-running model with the new figures.

5 Permitting and construction of the plant itself.

6 Commissioning of the plant, then operating years 1 and 2 at 60% and 85% of the capacity of the plant.

7 Review at end of operating year 3 against full-capacity output of 16,330 tonnes and against operating cash of **$7.4 million**.

FINANCIAL CLOSE GATE

Step 4 is the gate, given **no construction contract is signed** until quoted capital and signed offtake keep net present value of the project positive.

Section 12 | Safeguards and social due diligence

We have done **no environmental or social review** for this assessment, and the items that follow **must be completed and cleared** before construction commitment is made.

- **Environmental review** of chosen site, covering air emissions from pyrolysis units, noise, water use and handling of ash or fines, with the permits these items require.
- **Feedstock sourcing review** confirming where the residue comes from and that the residue meets feedstock rules of the chosen registry.
- **Worker health and safety plan** covering risks of fire and dust at the plant itself and in storage of feedstock.
- **Land and site status** confirming ownership or lease of site and any rights of access or of power line associated with the site.
- **Stakeholder consultation** with nearby communities and land agencies, with grievance mechanism for complaints raised by them.
- **Environmental and social management plan** setting out mitigation measures, with semiannual reports of monitoring during construction and annual reports of monitoring once the plant itself is in operation.

Monitoring schedule follows the arrangement in the ADB Southern Thailand Waste-to-Energy Project report, where borrower reports semiannually during construction and annually thereafter.

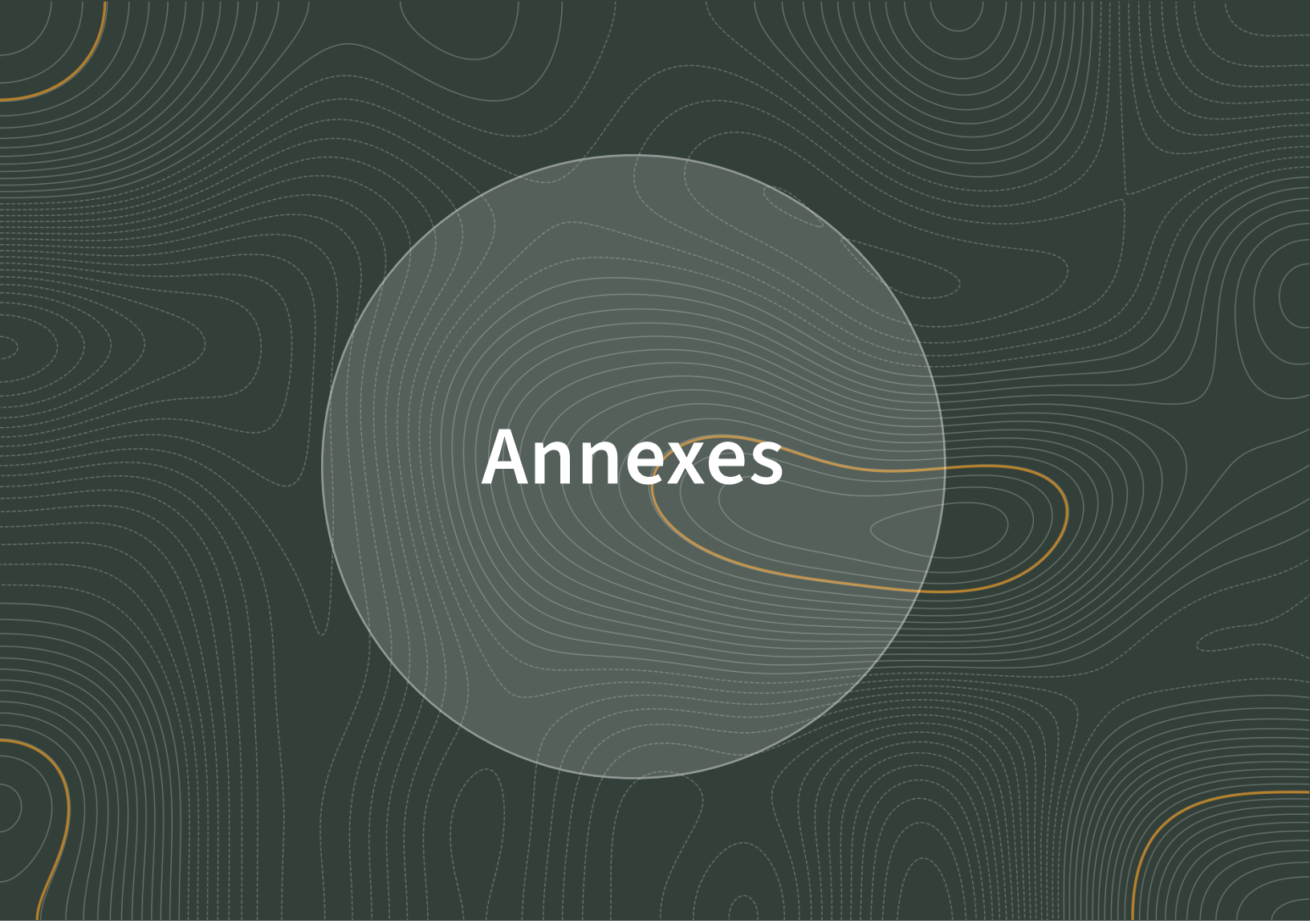
Section 13 | Recommendation and conditions

Approve the project to development stage, with **$19.1 million** construction commitment being subject to **five conditions**.

1. Signed offtake of biochar for most of the output planned for the plant at **$500 a tonne or more**, with floor price **no lower than $192**.
2. Signed credit offtake with the credit buyers for **at least 5 years**, with registry certification in place before start-up of the plant itself.
3. Lab tests on the biochar itself confirm credited CO2 levels near **2.5 tonnes** per tonne of biochar.
4. Feedstock contracts lock delivered cost of feedstock at or near **$55 a dry tonne**.
5. Fixed-price quotes keep capital cost of the plant **at or below $28.1 million**.

IF A CONDITION FAILS

Re-run this assessment with the new figures before committing the capital to the project.



Assumptions, model and sources

A	Assumption register	27
B	Model, method and checks	29
C	Sources	31

Annex A | Assumption register

Every input of the model, with status of the input and exact place it comes from.

FIGURE 12



Figure 12. Entries in model input file, inputs.csv, by status (with sensitivity bounds counted as separate entries).

Input	Value	Status	Source and exact location
Dry feedstock processed	45,360 t a year	Sourced	Sessions et al. 2019, p. 177, Section 2.2.1
Biochar yield, thermal pyrolysis	36%	Sourced	Sessions et al. 2019, p. 178, Section 2.2.2 and Table 2
Capital, reference design	$19.1 million	Sourced	Sessions et al. 2019, p. 180, Table 5 (Worden, thermal, Scenario 2)
Capital range	$17.0M to $28.1M	Sourced	Sessions et al. 2019, p. 180, Table 5 (lowest and highest of 16 designs)
Delivered feedstock cost	$55 a dry t	Sourced	Sessions et al. 2019, p. 181, Section 3.5 (Worden)
Feedstock cost range	$43 to $135 a dry t	Sourced	Sessions et al. 2019, p. 179, Section 3.1
Direct production cost	$207 per t biochar	Sourced	Sessions et al. 2019, p. 182, Table 9 (Worden, thermal, Scenario 2)
Production cost range, thermal	$184 to $254 per t	Sourced	Sessions et al. 2019, p. 182, Table 9
Operating life	20 years	Sourced	Sessions et al. 2019, p. 182, Table 9 header
Salvage value	20% of capital	Sourced	Sessions et al. 2019, p. 182, Table 9 header
Staff	26.5 FTE	Sourced	Sessions et al. 2019, p. 181, Table 6
Credit price	$148 per t CO2	Sourced	S&P Global, Oct. 2025, as reported by Carbon Credits, Dec. 3, 2025
Biochar sale price	$500 per t (tested $250 to $700)	Assumption	Context: 2014 survey wholesale average $1,500 per t, Sessions et al. 2019, p. 183
Credit price range	$80 to $220 per t CO2	Assumption	None

CO2 credited	2.5 t per t biochar (tested 1.8 to 3.0)	Assumption	To be replaced by lab test and registry method
Credit fees	15% of credit revenue	Assumption	None
Start-up	60%, 85%, then 100%	Assumption	None
Debt share and rate	60% at 7.5% nominal	Assumption	To be replaced by lender term sheet
Equity share and cost	40% at 15.0% nominal	Assumption	To be replaced by sponsor hurdle rate
Weighted average cost of capital	10.5% nominal, pre-tax	Computed	From the two rows above
Prices and costs	Flat in nominal terms, pre-tax	Assumption	None

Annex B | Model, method and checks

Every number in this report comes from **one run of Python model**, model.py, which reads one input file, inputs.csv. Changing an input and rerunning the model **regenerates every figure** in the report.

File	Contents
inputs.csv	Every input: value, unit, status, source and exact page, table or section
designs.csv	Capital (p. 180, Table 5) and total cost per tonne (p. 182, Table 9) for the 16 designs in Sessions et al. 2019
model.py	The model: cash flow, return measures, sensitivity, scenarios, break-even values, price curve, checks
outputs/results.md and results.json	Results of the run, including every year of the cash flow

Method

1. We compare the project with without-project case in which no plant is built, and we count only the cash flows that the plant itself creates.
2. Year 0 is construction, when the capital is spent on building the plant. Years 1 to 20 are years of operation.
3. Each year, biochar output is dry feedstock times yield times capacity use (60% in year 1, 85% in year 2, 100% after year 2).
4. Revenue is biochar tonnes times biochar price, plus biochar tonnes times CO2 credited per tonne times the credit price, less share of the credit fees.
5. Operating cost is feedstock tonnes times delivered feedstock cost, plus biochar tonnes times the direct cost of production of biochar.
6. Salvage value, which is **20% of capital**, is added in year 20 to the cash flow.
7. Net benefit each year is revenue minus operating cost plus salvage, and all flows are pre-tax and in flat nominal dollars, as we described above.

8 Weighted average cost of capital is debt share times the debt rate plus equity share times the cost of equity.

9 Financial internal rate of return is the discount rate at which net benefits sum to zero in present value, while net present value is the sum of net benefits discounted at the cost of capital. Payback is the time until cumulative net benefit turns positive.

10 Standard tests change revenue, biochar price, credit price, operating cost or capital by 10%, and delay operation one year with 10% more capital. Switching value is the percentage change in one input that brings net present value to zero, which we find through numerical root search.

11 Range tests set one input at a time to its low and high value, while scenarios change several inputs together. We find break-even values through root search on net present value of the project.

Checks against the source study

Check	Model	Sessions et al. 2019	Location
Biochar output at full capacity	16,329.6 t	about 16,330 t	p. 184, Section 4
Direct production plus feedstock cost per t biochar	$359.8	$359	p. 182, Table 9

Capital charge of the study of $86 per tonne (p. 182, Table 9) is not reproduced, given the model values capital through the cash flow and not through a capital charge.

Run on October 10, 2026. Results: cost of capital 10.5%; return 33.8%; net present value $38,936,501; payback 3.1 years.

Annex C | Sources

Sources of data.

Data sources

- Can biochar link forest restoration with commercial agriculture?, J. Sessions et al., Biomass and Bioenergy 123: 175 to 185, 2019
- Estimating profitability of two biochar production scenarios: slow pyrolysis vs fast pyrolysis, Iowa State University Bioeconomy Institute (T. Brown, M. Wright, R. Brown, 2010)
- Biochar Carbon Credits in 2025: Stable Prices Amid Weakening Demand, Carbon Credits, Dec. 3, 2025 (reporting S&P Global prices)
- Biochar carbon removal market snapshot 2025, CDR.fyi, Sep. 9, 2025
- Carbon removal prices 2026 to 2040, Senken, Aug. 20, 2026
- Process Feasibility Analysis of Waste Biomass Valorization to Biochar and Bio-Oil via Slow and Fast Pyrolysis, NREL, Energy and Fuels, 2025